

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Faculty of Technology & Engineering
Devang Patel Institute of Advance Technology and Research (DEPSTAR)

CSUE301 – : Big Data Analytics
Practical File – ODD Semester 2026-27

StudentID: 24DIT050
Student Name: Shalvi Patel

Practical 9

1. Problem Definition

TrendPulse Social Media Analytics monitors social media activity across platforms such as Twitter/X, Instagram, Facebook, and LinkedIn. Large volumes of posts contain hashtags related to technology, sports, entertainment and other topics.

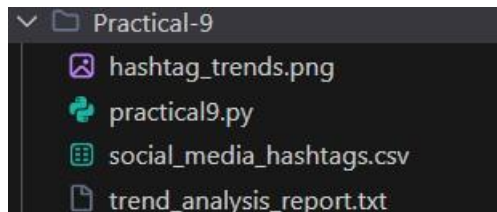
The objective of this practical is to use **PySpark** to process hashtag data and identify popular and trending topics.

The practical performs hashtag grouping, frequency counting, popularity ranking, daily trend analysis, regional analysis, category analysis and visualization.

Note: The dataset used in this practical is a generated/sample dataset for educational analysis.

PART A — Practical Execution

Step 1: Creating Folder and files



Step 2: Creating Spark Session

```
=====
TASK 1: CREATING SPARK SESSION
=====
WARNING: Using incubator modules: jdk.incubator.vector
26/09/27 17:08:49 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset. -see
https://wiki.apache.org/confluence/display/HADOOP2/WindowsProblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/09/27 17:08:50 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark Version   : 4.2.0
Spark App Name  : Social-Media-Trend-Detection
Master URL      : local[*]
```

Step 3: Loading Social Media Hashtag Dataset

```
=====
TASK 2: LOADING SOCIAL MEDIA HASHTAG DATASET
=====
```

```
Dataset Path      : C:\Users\ADMIN\OneDrive\Desktop\BDA\Practical-9\social_media_hashtags.csv
Dataset loaded successfully.
Total Number of Records: 650
```

```
First 10 Records:
```

```
+-----+-----+-----+-----+
|Timestamp|Platform|Hashtag      |Region      |
+-----+-----+-----+-----+
|2026-03-04|LinkedIn|#Netflix     |Latin America|
|2026-03-04|Facebook|#CloudComputing|North America|
|2026-03-01|Instagram|#Netflix     |Asia-Pacific |
|2026-03-04|Facebook|#CloudComputing|Middle East  |
|2026-03-01|LinkedIn|#Tech        |Middle East  |
|2026-03-01|LinkedIn|#Movies      |Europe       |
|2026-03-01|LinkedIn|#Hollywood   |Europe       |
|2026-03-04|LinkedIn|#Netflix     |North America|
|2026-03-04|LinkedIn|#AI          |North America|
|2026-03-05|Twitter/X|#AI          |North America|
+-----+-----+-----+-----+
```

```
only showing top 10 rows
```

Step 4: Displaying Hashtag Records

```
=====
TASK 3: DISPLAYING HASHTAG RECORDS
=====
```

```
Sample Hashtag Records (First 10 rows):
```

```
+-----+-----+-----+-----+
|Timestamp|Platform|Hashtag      |Region      |
+-----+-----+-----+-----+
|2026-03-04|LinkedIn|#Netflix     |Latin America|
|2026-03-04|Facebook|#CloudComputing|North America|
|2026-03-01|Instagram|#Netflix     |Asia-Pacific |
|2026-03-04|Facebook|#CloudComputing|Middle East  |
|2026-03-01|LinkedIn|#Tech        |Middle East  |
|2026-03-01|LinkedIn|#Movies      |Europe       |
|2026-03-01|LinkedIn|#Hollywood   |Europe       |
|2026-03-04|LinkedIn|#Netflix     |North America|
|2026-03-04|LinkedIn|#AI          |North America|
|2026-03-05|Twitter/X|#AI          |North America|
+-----+-----+-----+-----+
```

```
only showing top 10 rows
```

Step 5: Validating Dataset Structure

```
=====
TASK 4: VALIDATING DATASET STRUCTURE
=====

DataFrame Schema:
root
|-- Timestamp: date (nullable = true)
|-- Platform: string (nullable = true)
|-- Hashtag: string (nullable = true)
|-- Region: string (nullable = true)

Number of Columns : 4
Number of Records : 650
Column Names      : Timestamp, Platform, Hashtag, Region
```

Step 6: Grouping Hashtags

```
=====
TASK 5: GROUPING HASHTAGS
=====

Hashtag Occurrences Grouped (Top 15):
+-----+-----+
|Hashtag      |count|
+-----+-----+
|#AI          |75   |
|#Cricket     |68   |
|#MachineLearning|58   |
|#Netflix     |52   |
|#DataScience |45   |
|#Football    |42   |
|#Gaming       |38   |
|#Tech        |35   |
|#IPL         |34   |
|#CloudComputing|32   |
|#CyberSecurity|28   |
|#Movies      |26   |
|#Music       |24   |
|#Sports      |22   |
|#Bollywood   |20   |
+-----+-----+
only showing top 15 rows
```

Step 7: Counting Hashtag Occurrences

TASK 6: COUNTING HASHTAG OCCURRENCES

Top 10 Most Frequent Hashtags:

Hashtag	count
#AI	75
#Cricket	68
#MachineLearning	58
#Netflix	52
#DataScience	45
#Football	42
#Gaming	38
#Tech	35
#IPL	34
#CloudComputing	32

Step 8: Sorting Hashtags by Popularity

TASK 7: SORTING HASHTAGS BY POPULARITY

Hashtags Sorted by Popularity with Dynamic Rank:

Rank	Hashtag	Occurrence_Count
1	#AI	75
2	#Cricket	68
3	#MachineLearning	58
4	#Netflix	52
5	#DataScience	45
6	#Football	42
7	#Gaming	38
8	#Tech	35
9	#IPL	34
10	#CloudComputing	32
11	#CyberSecurity	28
12	#Movies	26
13	#Music	24
14	#Sports	22
15	#Bollywood	20

only showing top 15 rows

Step 9: Identifying Top Trending Hashtags

TASK 8: IDENTIFYING TOP TRENDING HASHTAGS

Top 5 Trending Hashtags:

Rank	Hashtag	Occurrence_Count
1	#AI	75
2	#Cricket	68
3	#MachineLearning	58
4	#Netflix	52
5	#DataScience	45

[>>>] Top Trending Hashtag: #AI with 75 posts

Step 10: Supplementary 1 — Top 3 Trending Hashtags

SUPPLEMENTARY 1: TOP 3 TRENDING HASHTAGS

Rank	Hashtag	Occurrence_Count
1	#AI	75
2	#Cricket	68
3	#MachineLearning	58

Top Trending Hashtag:
#AI (Total Occurrences: 75)

Step 11: Supplementary 2 Daily Trend Comparison

```
=====
SUPPLEMENTARY 2: DAILY TREND COMPARISON
=====
Comparing Hashtag Trends Across Multiple Days...

Daily Hashtag Counts (Sample 12 records):
+-----+-----+-----+
|Timestamp|Hashtag      |count|
+-----+-----+-----+
|2026-03-01|#AI          |16|
|2026-03-01|#Netflix     |12|
|2026-03-01|#Cricket      |11|
|2026-03-01|#DataScience  |10|
|2026-03-01|#Movies       |9|
|2026-03-01|#IPL          |8|
|2026-03-01|#CloudComputing|7|
|2026-03-01|#Bollywood    |7|
|2026-03-01|#Tech         |7|
|2026-03-01|#Gaming       |6|
|2026-03-01|#Football     |6|
|2026-03-01|#MachineLearning|6|
+-----+-----+-----+
only showing top 12 rows

Most Popular Hashtag Per Day:
+-----+-----+-----+
|Timestamp|Daily_Top_Hashtag|Daily_Count|
+-----+-----+-----+
|2026-03-01|#AI          |16|
|2026-03-02|#AI          |17|
|2026-03-03|#Cricket      |16|
|2026-03-04|#AI          |19|
|2026-03-05|#Cricket      |14|
+-----+-----+-----+
```

Step 12: Supplementary 3 Regional Hashtag Analysis

```
=====
SUPPLEMENTARY 3: REGIONAL HASHTAG ANALYSIS
=====

Total Posts Per Region:
+-----+-----+
|Region      |Total_Posts|
+-----+-----+
|Asia-Pacific|145         |
|North America|142        |
|Middle East |122         |
|Europe       |121         |
|Latin America|120        |
+-----+-----+

Top Trending Hashtag Per Region:
+-----+-----+-----+
|Region      |Top_Hashtag|Post_Count|
+-----+-----+-----+
|Asia-Pacific|#Cricket    |20         |
|Latin America|#MachineLearning|20         |
|North America|#AI         |20         |
|Middle East  |#AI         |16         |
|Europe       |#MachineLearning|15         |
+-----+-----+-----+
```

Step 13: Task 10 / Supplementary 4 — User Interest and Hashtag Categories

```
=====
TASK 10 & SUPPLEMENTARY 4: USER INTEREST & HASHTAG CATEGORIES
=====

User Interest Distribution by Category:
+-----+-----+
|Category      |Number_of_Posts|
+-----+-----+
|Technology     |273             |
|Sports         |199             |
|Entertainment  |178             |
+-----+-----+

[>>>] Most Popular Category: Technology with 273 posts

Platform Distribution:
+-----+-----+
|Platform |Total_Posts|
+-----+-----+
|Facebook |172         |
|Instagram|165         |
|LinkedIn |160         |
|Twitter/X|153         |
+-----+-----+
```


Step 14: Supplementary 5 Trend Visualization

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SUPPLEMENTARY 5: TREND VISUALIZATION

===== Creating

Matplotlib Bar Chart for Top 10 Trending Hashtags...

Chart successfully saved to: C:\Users\ADMIN\OneDrive\Desktop\BDA\Practical-9\hashtag_trends.png

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BUSINESS RECOMMENDATIONS

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[DISCLAIMER]: Educational analysis based on generated/sample social media data.

1. Campaign Targeting : Allocate top-tier marketing budgets toward Technology and Sports hashtags.
2. Platform Strategy : Cross-post top tech trends on LinkedIn and fan-driven sports topics on Twitter/X.
3. Emerging Topics : Monitor real-time daily trend shifts to detect breakout virality early.
4. Geo-Targeted Content : Tailor campaigns based on regional preferences (e.g. cricket in Asia-Pacific).
5. Community Engagement : Create branded interactive polls around #AI, #Cricket, and #Netflix.

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FINAL SOCIAL MEDIA TREND ANALYSIS REPORT

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TRENDPULSE SOCIAL MEDIA TREND DETECTION REPORT

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1. DATASET DETAILS:

- Source File : social_media_hashtags.csv
- Total Records : 650
- Schema : Timestamp (string), Platform (string), Hashtag (string), Region (string)
- Data Nature : Generated / Sample dataset for educational evaluation

2. TOTAL RECORDS:

- Total Monitored Posts : 650

3. UNIQUE HASHTAGS:

- Total Distinct Hashtags : 18

4. PLATFORM DISTRIBUTION:

Platform	Total Posts
----------	-------------

----- Facebook

| 172

Instagram | 165

LinkedIn | 160

Twitter/X | 153

5. REGION DISTRIBUTION:

Region	Top Hashtag	Post Count
Asia-Pacific	#Cricket	20
Latin America	#MachineLearning	20
North America	#AI	20
Middle East	#AI	16
Europe	#MachineLearning	15

6. HASHTAG FREQUENCY ANALYSIS:

- Evaluated across all monitored interactions with dynamic grouping and frequency sorting.

7. TOP 10 HASHTAGS:

Rank	Hashtag	Occurrences
1	#AI	75
2	#Cricket	68
3	#MachineLearning	58
4	#Netflix	52
5	#DataScience	45
6	#Football	42
7	#Gaming	38
8	#Tech	35
9	#IPL	34
10	#CloudComputing	32

8. TOP 5 TRENDING HASHTAGS:

Rank	Hashtag	Occurrences
1	#AI	75
2	#Cricket	68
3	#MachineLearning	58
4	#Netflix	52
5	#DataScience	45

9. TOP 3 TRENDING HASHTAGS:

Rank	Hashtag	Occurrences
1	#AI	75
2	#Cricket	68
3	#MachineLearning	58

* Top Trending Hashtag : #AI (75 occurrences)

10. DAILY TREND ANALYSIS:

Date	Daily Top Hashtag	Post Count
2026-03-01	#AI	16
2026-03-02	#AI	17
2026-03-03	#Cricket	16
2026-03-04	#AI	19
2026-03-05	#Cricket	14

11. REGIONAL ANALYSIS:

Region	Top Hashtag	Post Count
Asia-Pacific	#Cricket	20
Latin America	#MachineLearning	20
North America	#AI	20
Middle East	#AI	16
Europe	#MachineLearning	15

12. CATEGORY ANALYSIS:

Category	Number of Posts
Technology	273
Sports	199
Entertainment	178

* Leading Category : Technology (273 posts)

13. USER INTEREST ANALYSIS:

- User attention is primarily concentrated in cutting-edge Technology topics (#AI, #MachineLearning), followed by high-energy Sports discussions (#Cricket, #Football) and Entertainment streaming (#Netflix).

14. BUSINESS RECOMMENDATIONS:

* DISCLAIMER: Educational analysis based on generated/sample social media data.

- Allocate promotional spending to high-affinity topics (#AI, #Cricket, #MachineLearning).
- Leverage multi-platform distribution based on user demographics (Tech -> LinkedIn, Sports -> Twitter/X).
- Monitor daily breakout hashtags to optimize real-time digital marketing responses.

15. CONCLUSION:

PySpark DataFrame transformations, grouping, and window functions offer massive scalability and low latency for parsing unstructured social interaction streams and extracting actionable trends.

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Key Questions / Analysis

Q1. Why is Apache Spark suitable for social media analytics?

Answer:

Apache Spark is suitable for social media analytics because it can process large amounts of data efficiently using distributed computing. Spark DataFrames provide fast operations such as filtering, grouping, aggregation and sorting, which are useful for analysing large social media datasets.

Q2. What is the significance of hashtag frequency?

Answer:

Hashtag frequency represents how many times a particular hashtag appears in the dataset. A higher frequency indicates that the topic is receiving more attention in the analyzed sample.

Q3. How can trend analysis support digital marketing campaigns?

Answer:

Trend analysis helps marketers identify popular topics and changing user interests. Businesses can use this information to create relevant content, plan campaigns and monitor emerging topics.

Q4. Why do businesses monitor social media trends?

Answer:

Businesses monitor social media trends to understand customer interests, identify emerging topics, observe discussions about products or services and improve their marketing strategies.

Q5. How can real-time analytics improve decision making?

Answer:

Real-time analytics allows organizations to detect changes in trends quickly. This can help businesses respond faster to emerging topics, customer interests and market changes.

Result

The Social Media Trend Detection practical was successfully implemented using PySpark.

The dataset contained 650 social media records and 18 distinct hashtags. Hashtag frequency, popularity ranking, top trends, daily trends, regional trends and category-based user interests were successfully analyzed.

The most frequently occurring hashtag in the sample dataset was #AI with 75 occurrences.

The Technology category had the highest number of posts with 273 posts. A visualization of the top 10 hashtags was also successfully generated using Matplotlib.

Conclusion

The practical demonstrated how **PySpark DataFrame operations and aggregation techniques** can be used for social media trend detection. Hashtags were grouped, counted and ranked according to their frequency.

Daily and regional analysis provided additional information about changing user interests. Category analysis showed the distribution of Technology, Sports and Entertainment topics.

The practical demonstrates that PySpark can be used to process and analyze large-scale social media datasets efficiently and extract useful trends for business analysis.

5. Github Link

Github Link: https://github.com/shalvi000009/24DIT050_BDA_Practical
